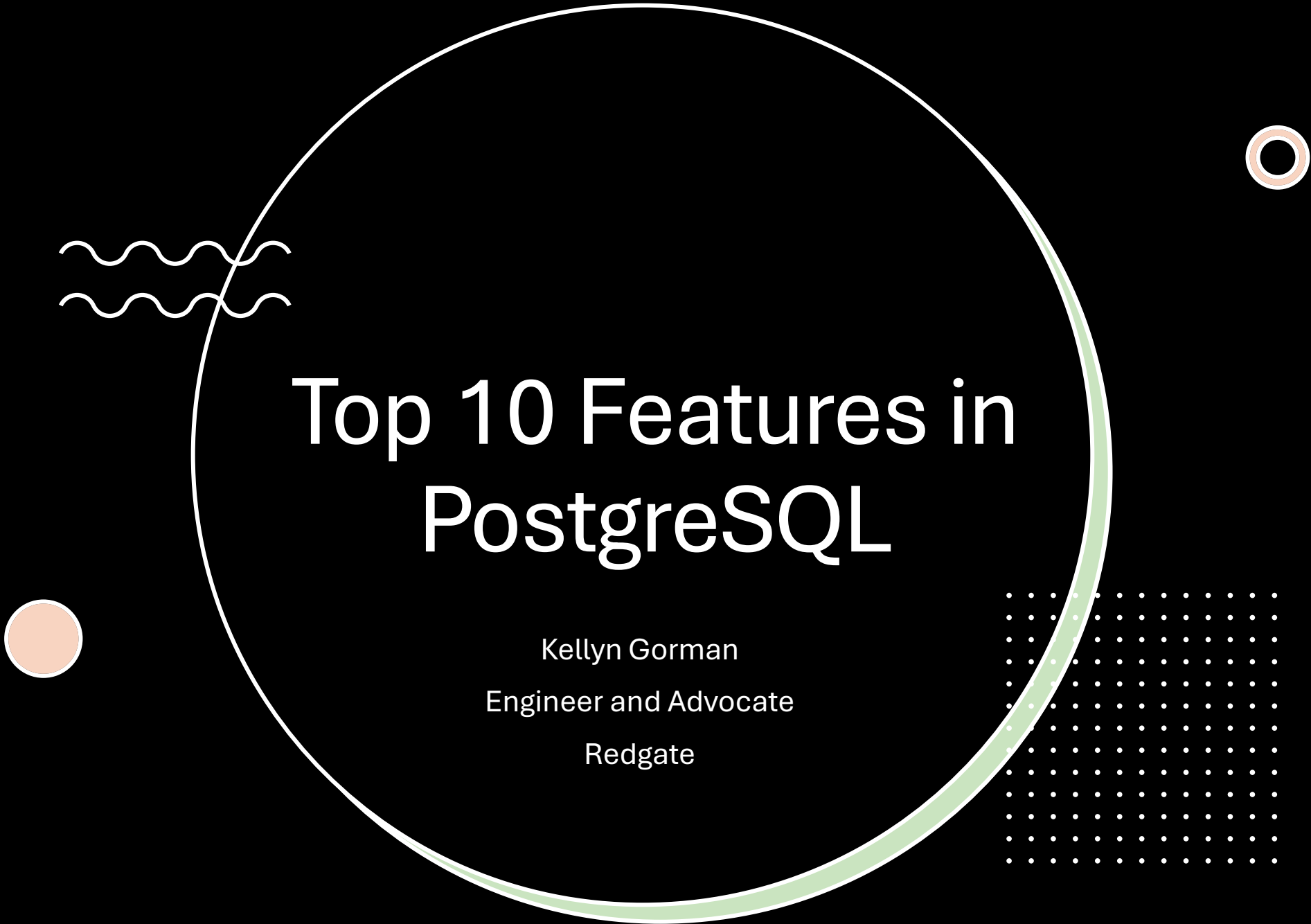




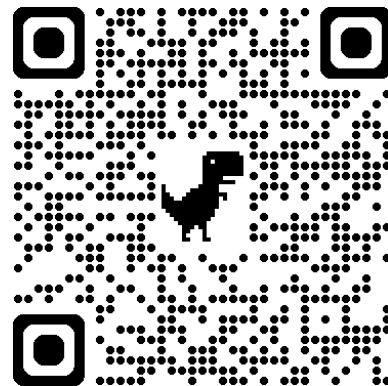
Top 10 Features in PostgreSQL

Kellyn Gorman
Engineer and Advocate
Redgate

Kellyn Gorman

Multiplatform and AI Advocate

<https://dbakevlar.com>

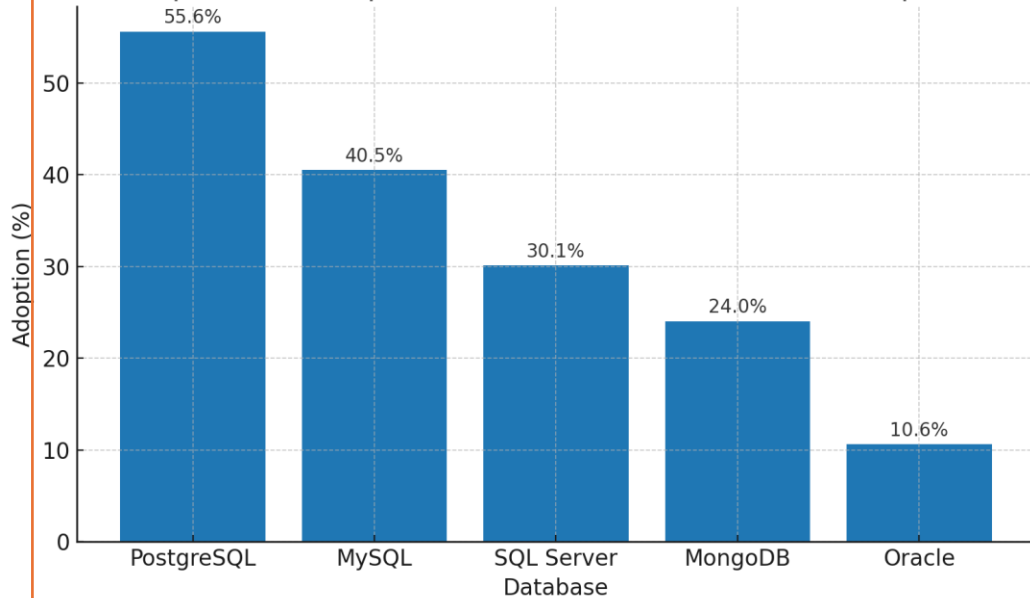




- Relational + Non-Relational Workloads
- Performance & Scalability
- Security & Extensibility
- Real-World Use Cases
- Demonstration
- Key Takeaways & Q&A

Why PostgreSQL Now?

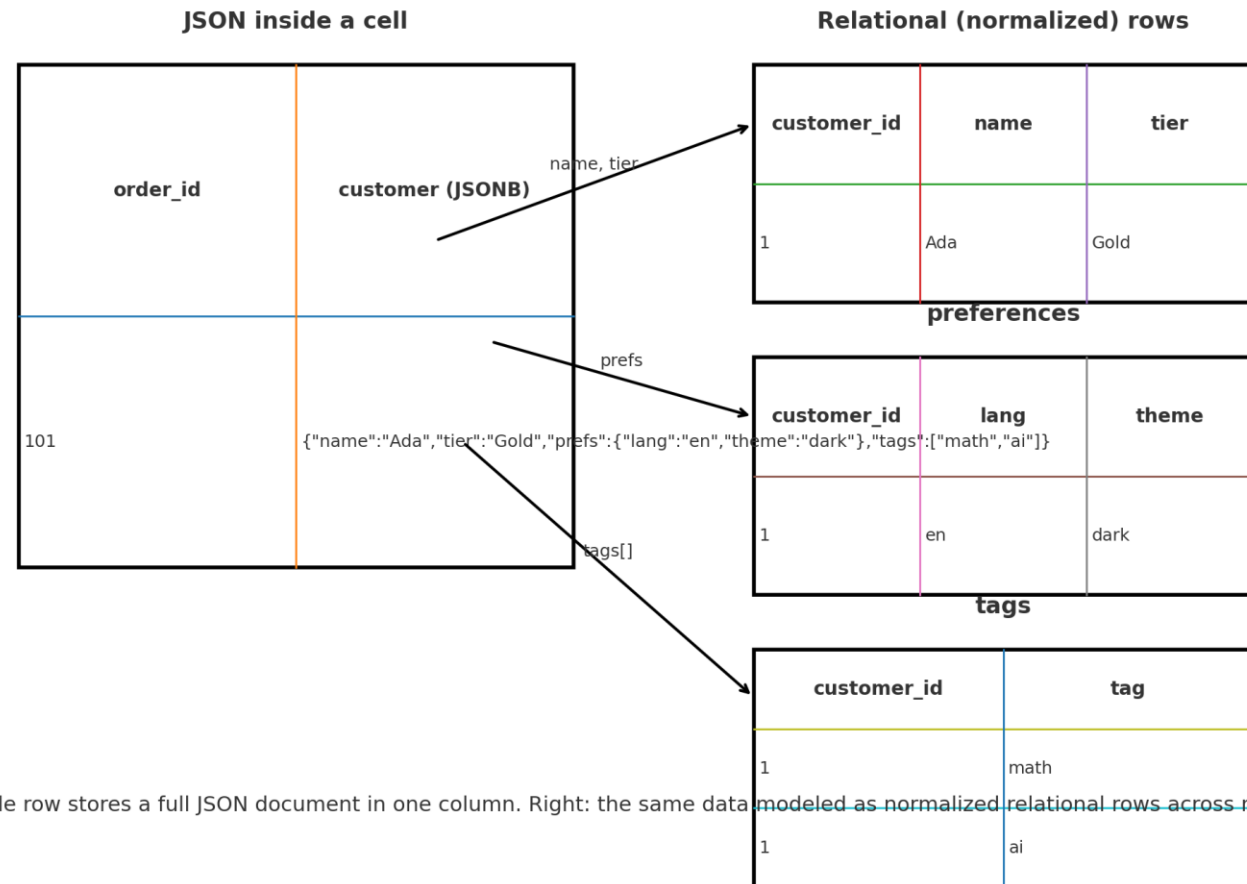
Database Adoption (All Respondents) — Stack Overflow Developer Survey 2025



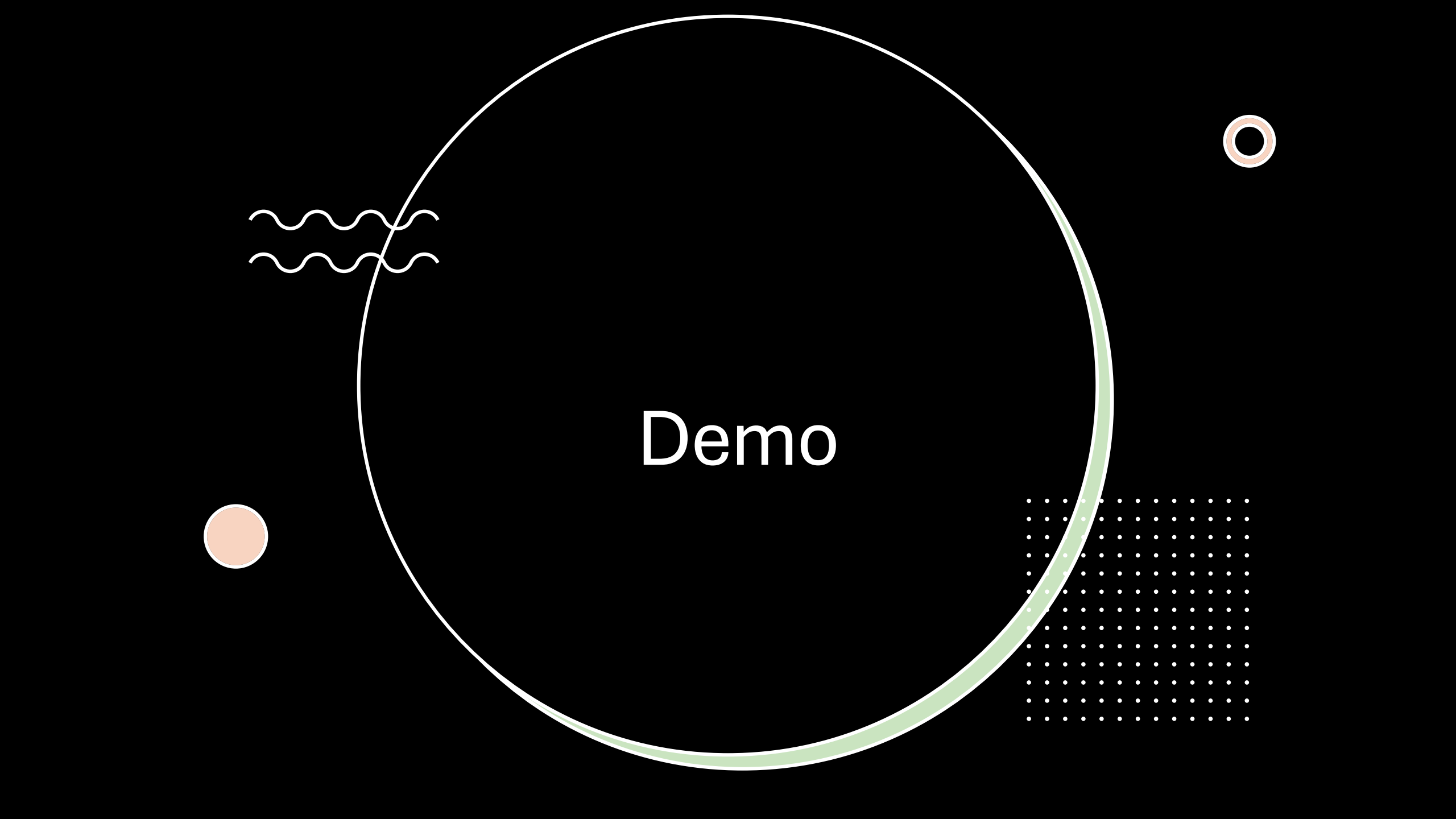
- Fastest-growing open-source RDBMS
- Cloud-native adoption (AWS, Azure, GCP)
- Ecosystem of extensions for AI, analytics, and beyond

1. Relational and Non-Relational Workloads

- JSON & JSONB
- Store/query semi-structured data
- Indexing on JSON fields for speed



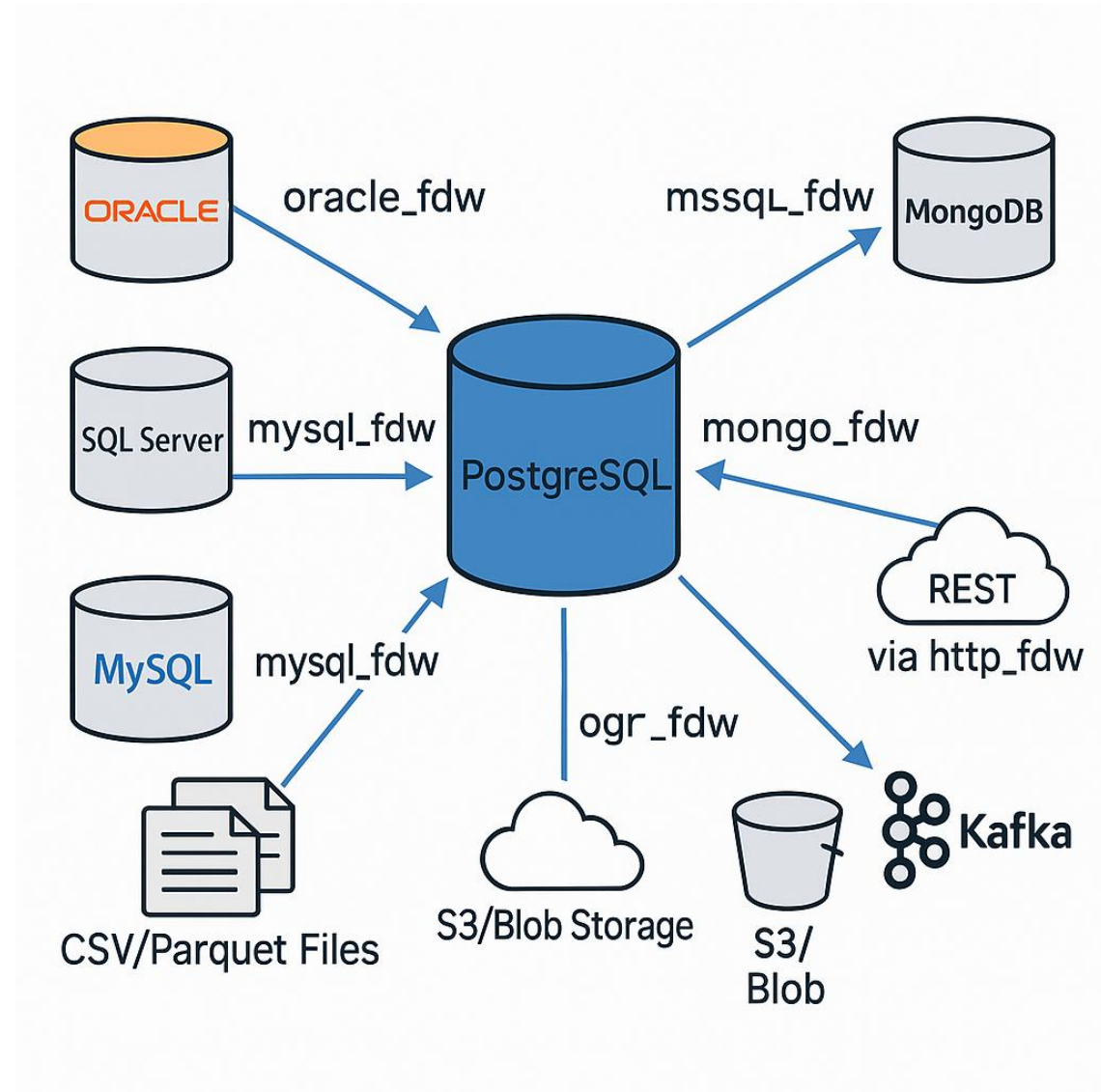
Left: a single row stores a full JSON document in one column. Right: the same data modeled as normalized relational rows across multiple tables.

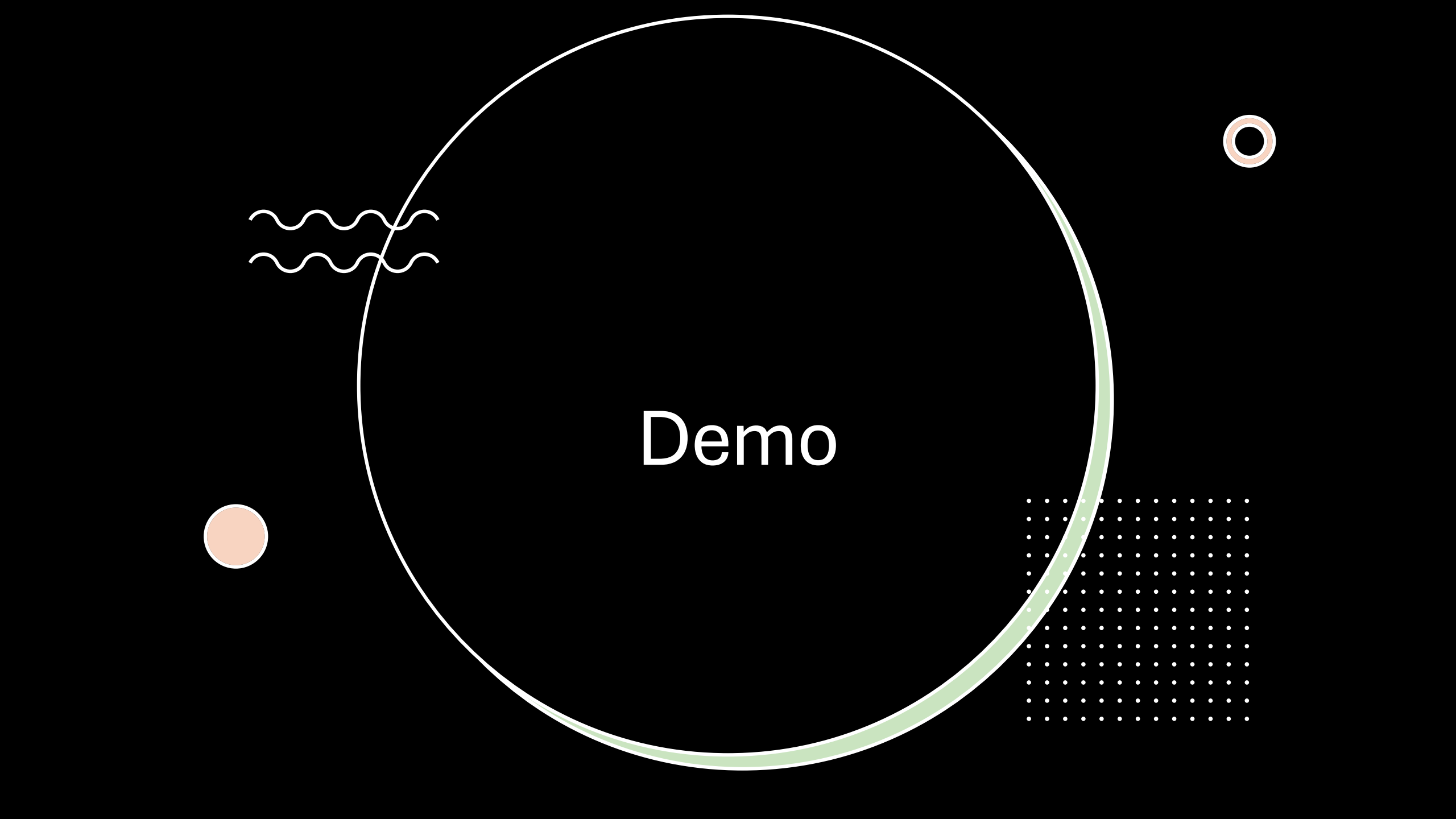
An abstract graphic design on a black background. A large, thin white circle is centered, with a thick light green ring following its outer edge. The word "Demo" is written in white serif font in the center. To the left of the circle, there are two white wavy lines and a solid light orange circle. To the right, there is a small double-lined light orange circle and a rectangular grid of small white dots.

Demo

2. Foreign Data Wrappers (FDWs)

- Query external sources: Oracle, MongoDB, APIs
- Unified SQL interface across multiple systems



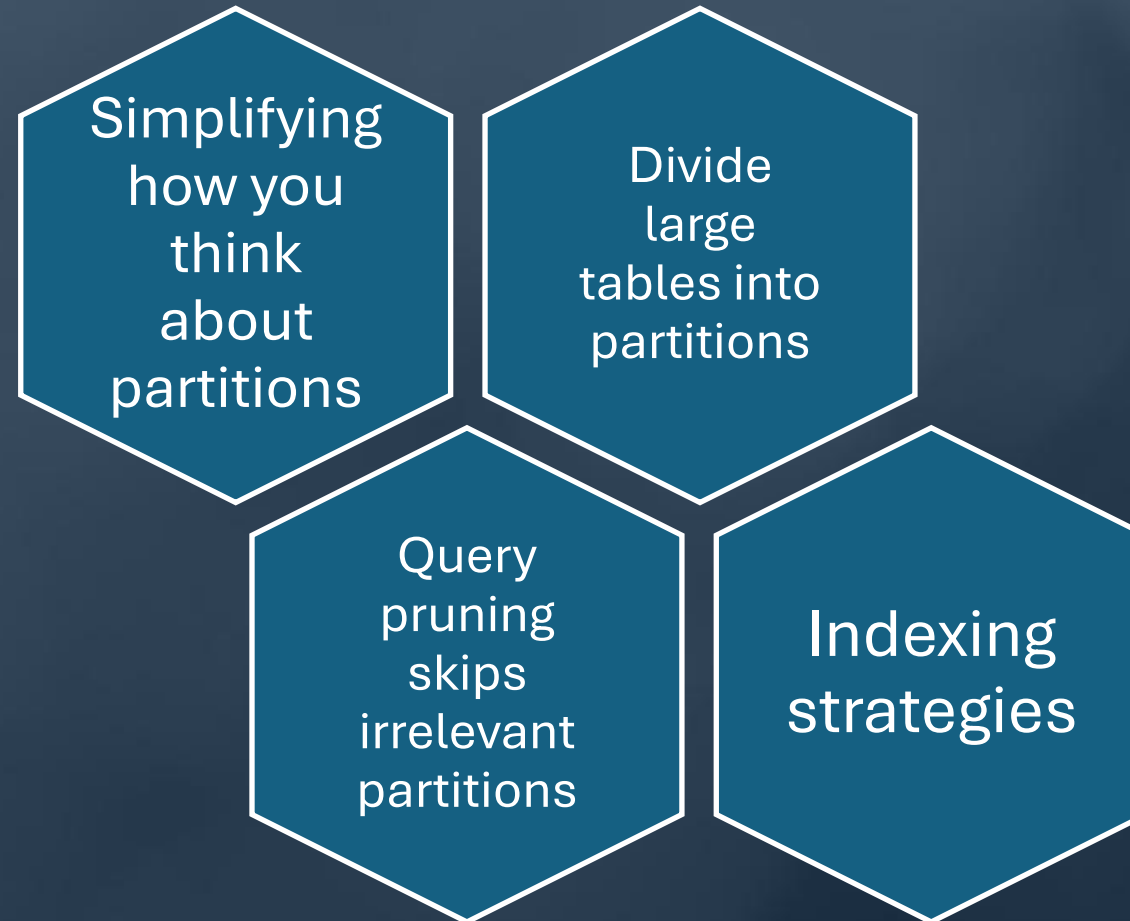
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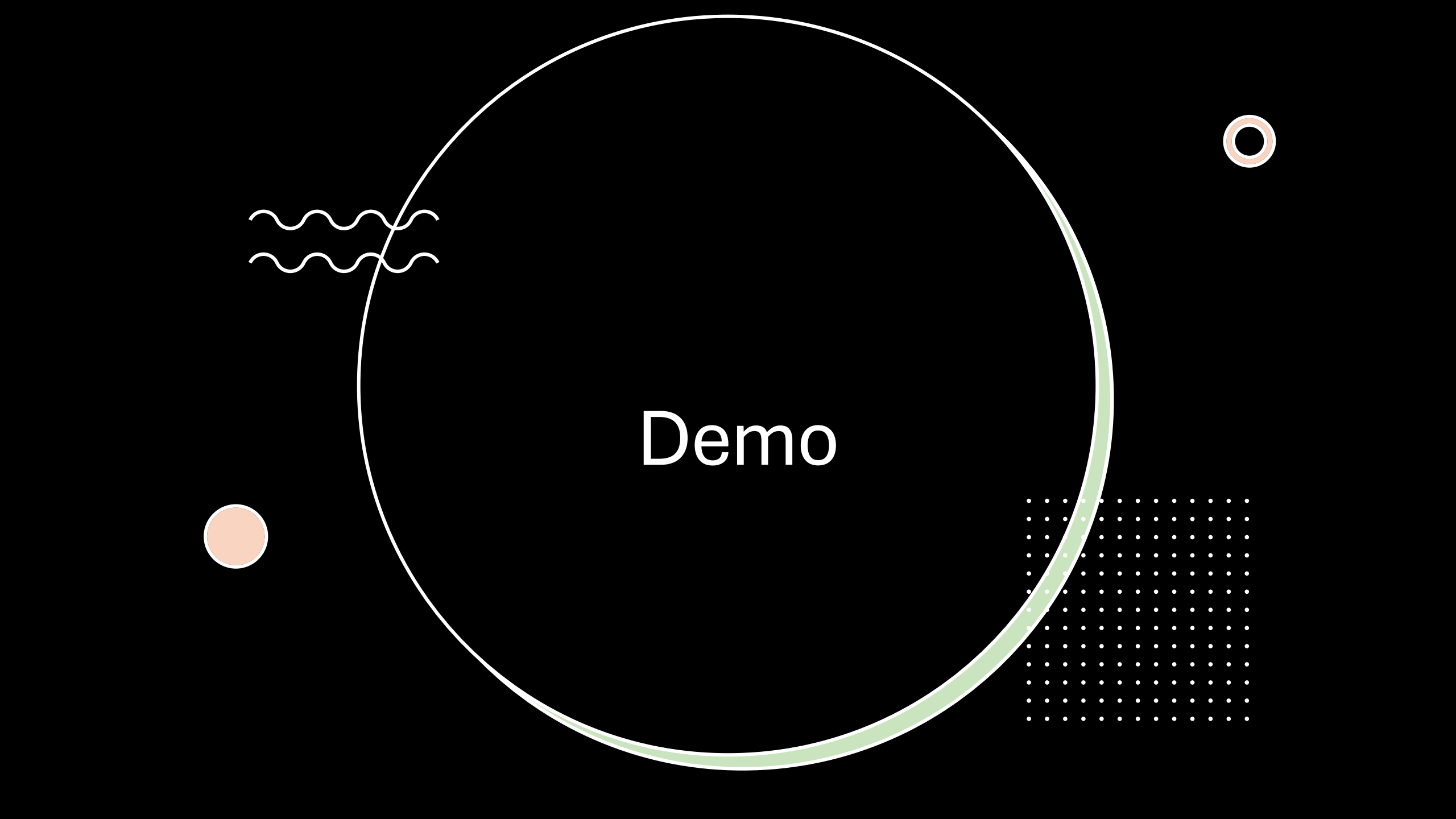
Demo

Performance and Scalability Overview

- Partitioning
- Parallel Queries
- Advanced Indexing

3. Partitioning



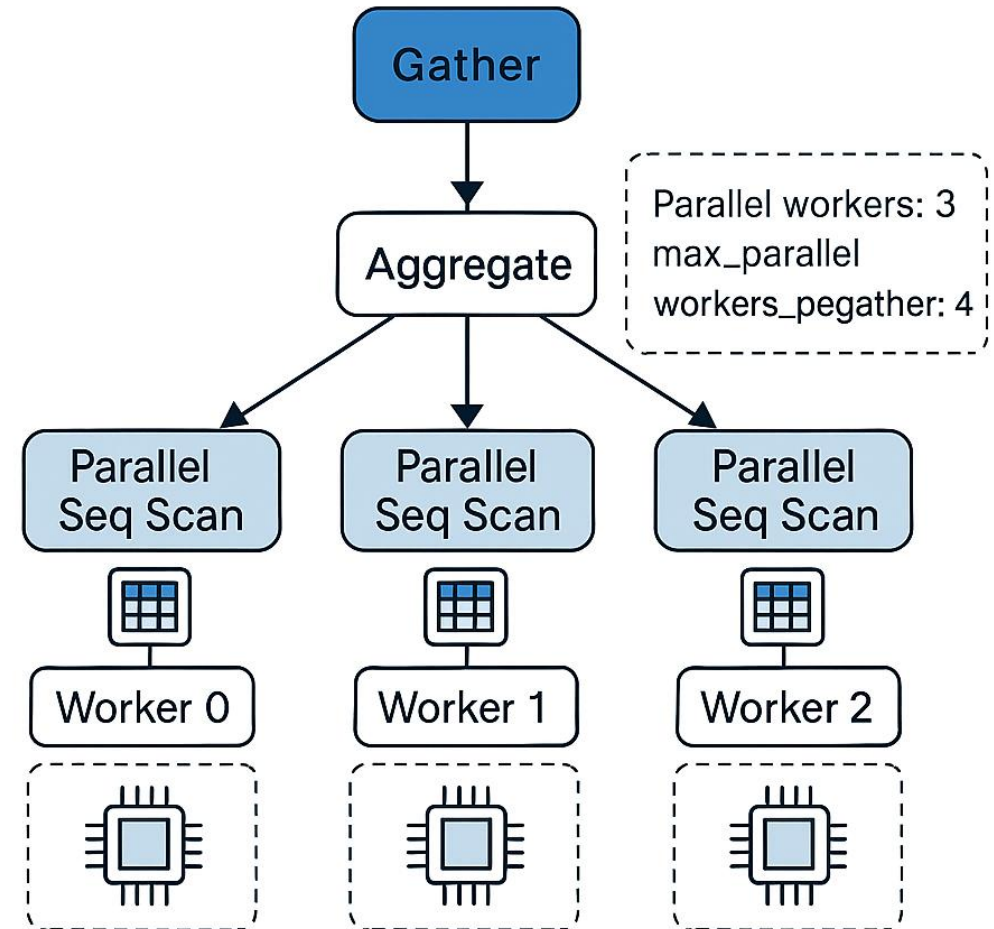
An abstract graphic design on a black background. A large, thin white circle is centered, with a thick light green ring following its outer edge. The word "Demo" is written in white serif font in the center. To the left of the circle, there are two white wavy lines and a solid light orange circle. To the right, there is a small double-lined light orange circle and a square grid of white dots. The green ring has a slight gradient and a shadow effect.

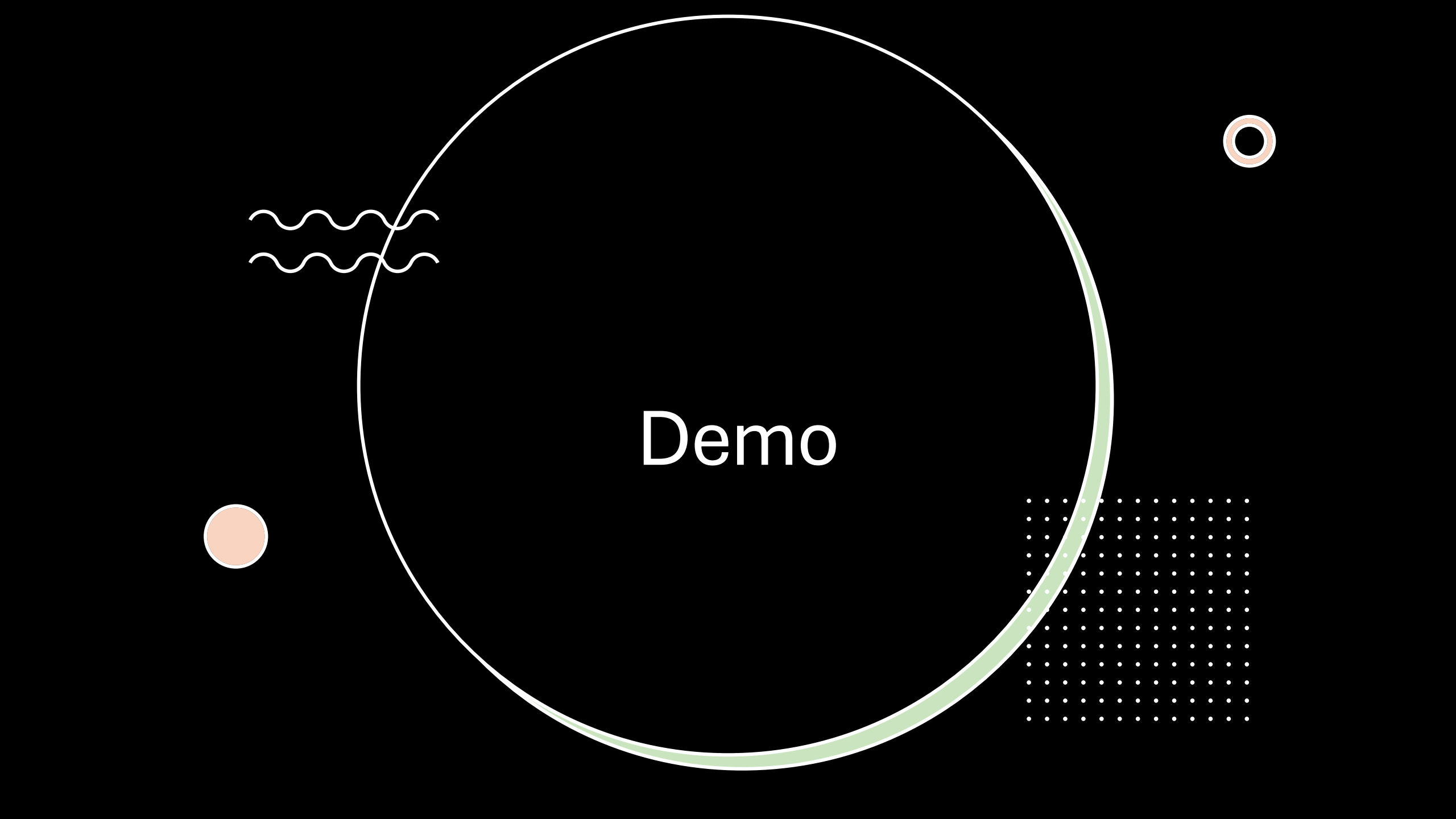
Demo

4. Parallel Queries

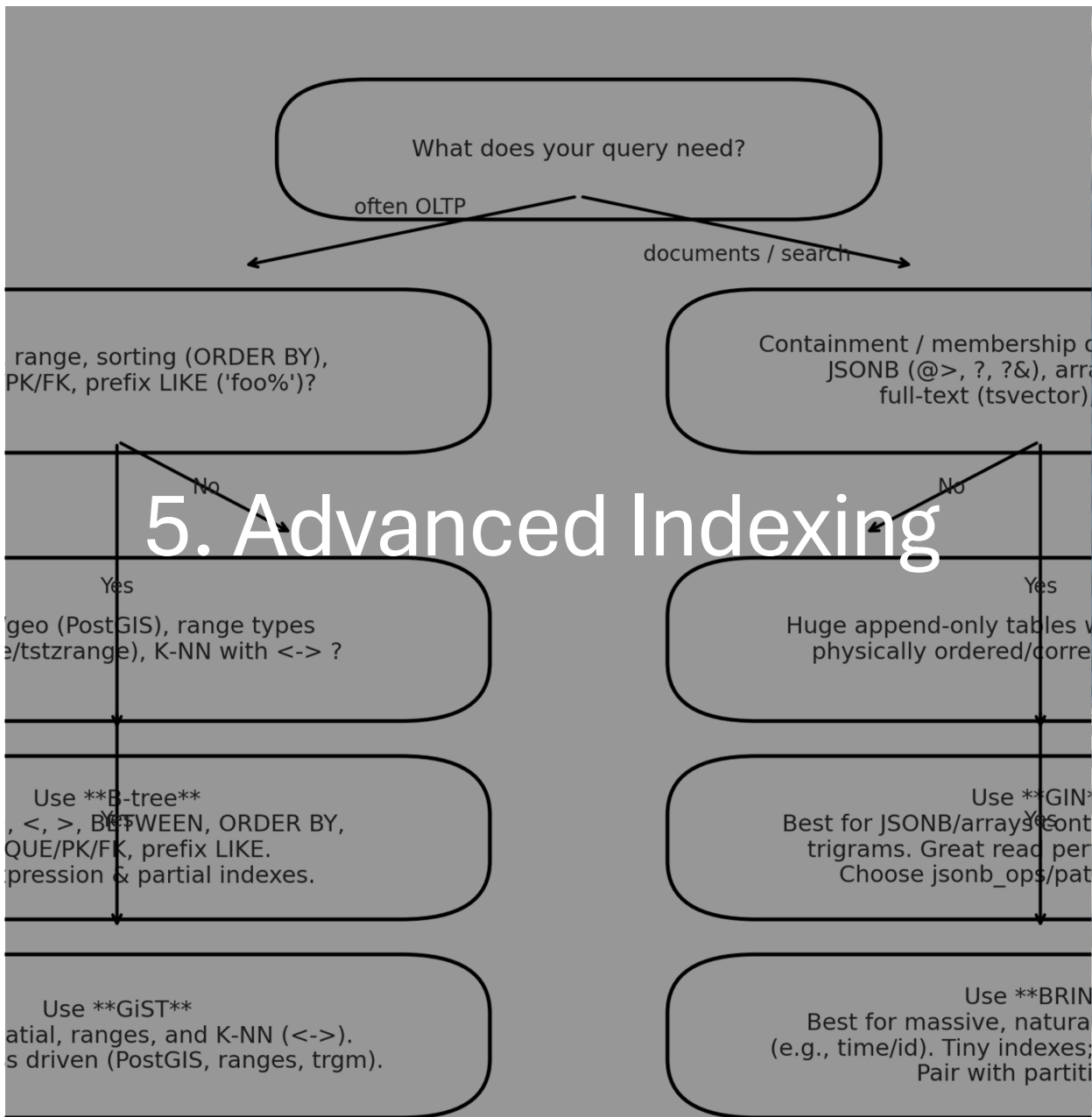
- Leverages multi-core CPUs
- Improves analytics query performance

Parallel Query Execution Plan



An abstract graphic design on a black background. A large white circle is centered, with a thick light green double-line border on its right side. The word "Demo" is written in white serif font in the center of the circle. To the left of the circle, there are two white wavy lines and a solid light orange circle. To the right, there is a small light orange circle with a white outline and a grid of small white dots.

Demo



5. Advanced Indexing

B-Tree:
traditional
queries

BRIN: large,
sequential
datasets

GIN: full-text
search,
JSONB

GiST:
geometric,
range queries

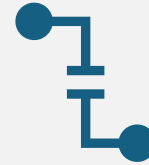
6. Security Best Practices



SSL/TLS FOR DATA IN
TRANSIT



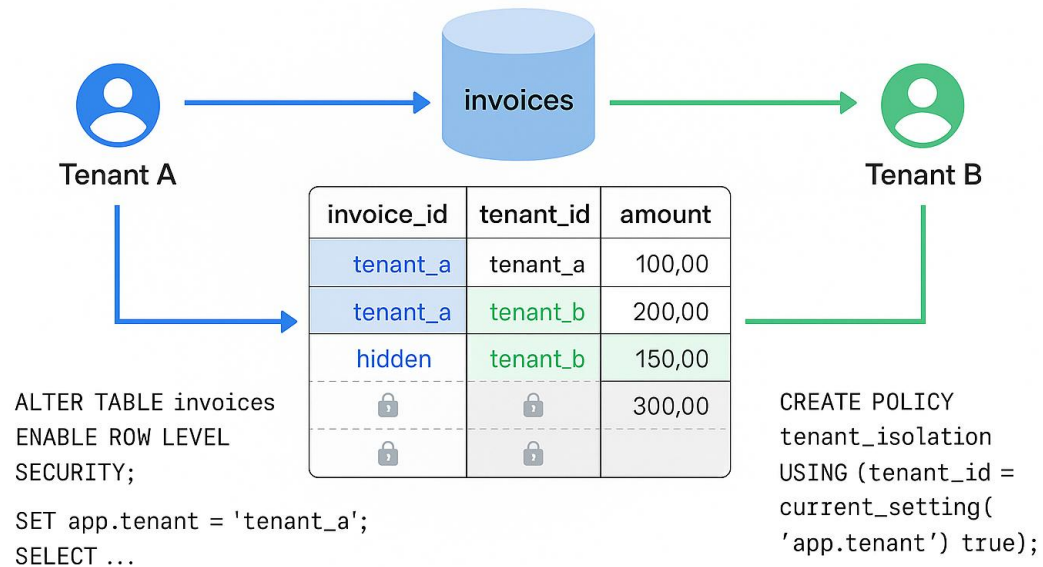
PGCRYPTO FOR
COLUMN ENCRYPTION



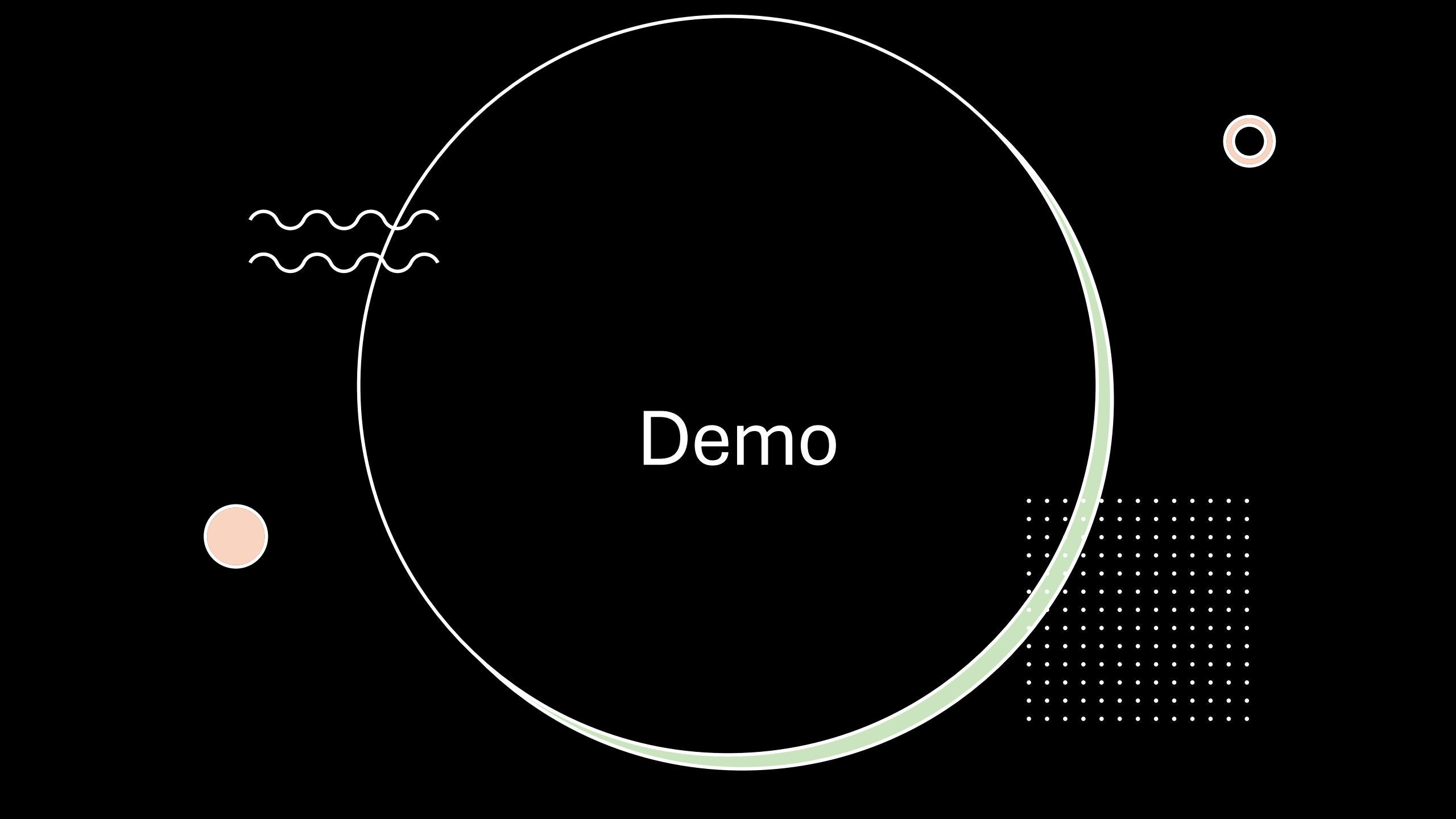
RLS FOR TENANT-
SPECIFIC ACCESS

7. Row-Level Security (RLS)

PostgreSQL Row-Level Security (RLS)

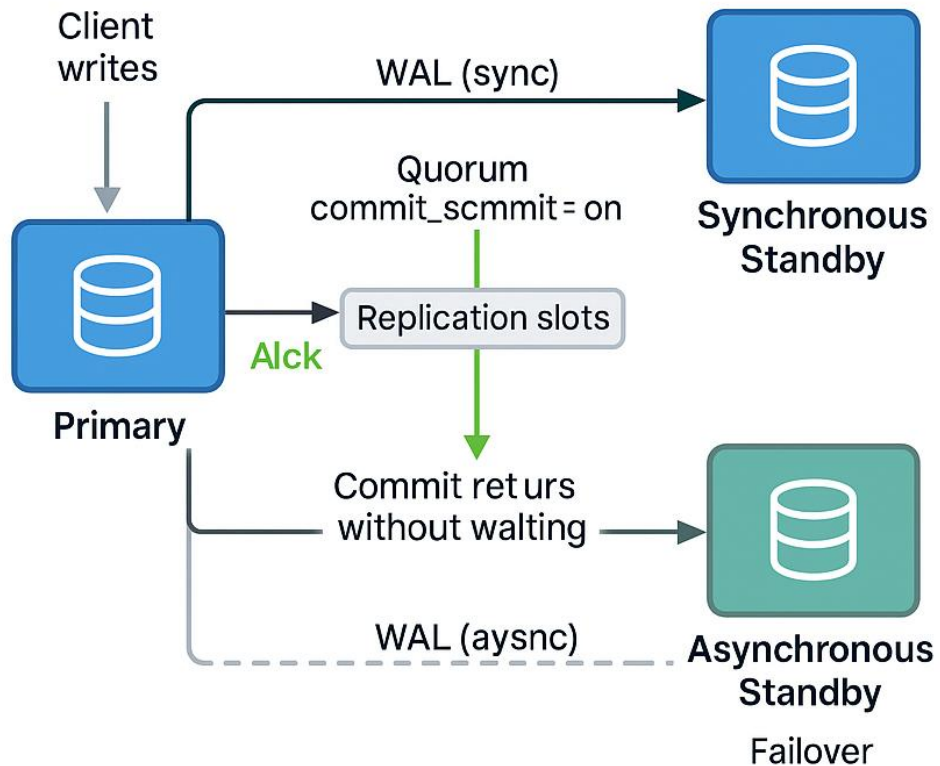


- Per-user or per-tenant visibility
- Policies defined in SQL

An abstract graphic design on a black background. A large, thin white circle is centered, with a thick light green ring following its outer edge. The word "Demo" is written in white serif font in the center. To the left of the circle, there are two white wavy lines and a solid light orange circle. To the right, there is a small double-lined light orange circle and a square grid of small white dots.

Demo

PostgreSQL Primary and Standby Replication



8. Replication for HA/DR

- Streaming replication
- Synchronous vs. asynchronous
- Logical replication for selective tables



9. Extensibility Superpowers

- PostGIS: spatial data
- pgvector: AI & embeddings
- Pg_stat_statements:
Performance data

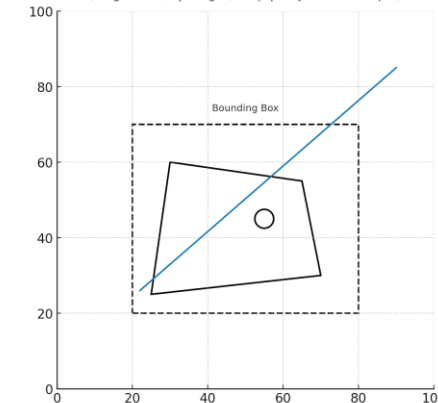


PostGIS

Spatial
functions,
GIS support

Mapping,
logistics, IoT
use cases

GIS Coordinates & Geometry
(Origin at 0,0 | X right, Y up | Projected example)



GIS Layers and their PostgreSQL/PostGIS Tables

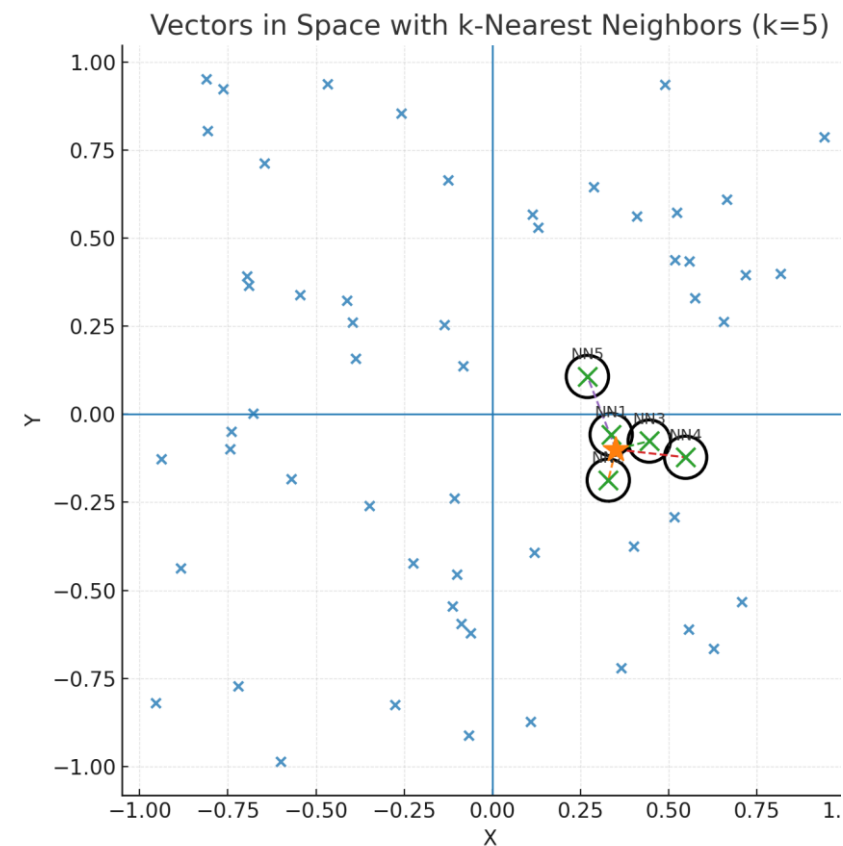
Points table: places geom: geometry(Point, 4326)
Lines table: roads geom: geometry(LineString, 3857)
Polygons table: parcels geom: geometry(Polygon, 3857)
Raster (optional) table: elevation raster: raster SRID=4326

Each layer is typically a table with a geometry/raster column and an SRID.
Use ST_AsText/ST_AsEWKB, ST_Transform to change SRID; add GIST/BRIN indexes on geom.

pgvector

Vector similarity
search for AI

Use with LLM
embeddings,
recommendations



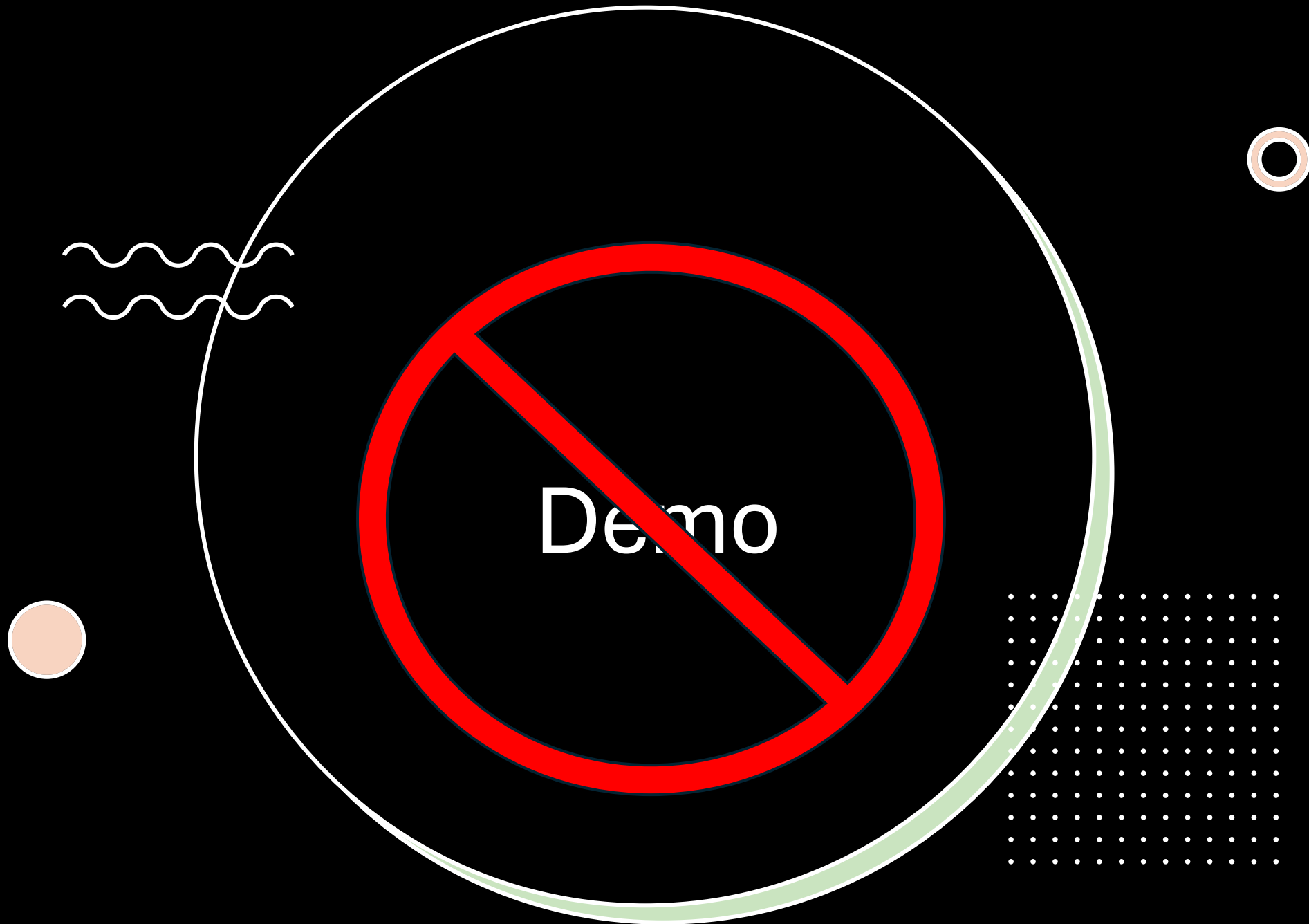
10. Custom Extensions



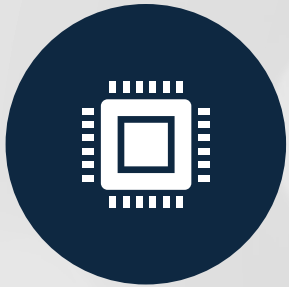
Write domain-specific
functions in C/SQL/PLpgSQL



Share via pgxn.org



Real-World Examples



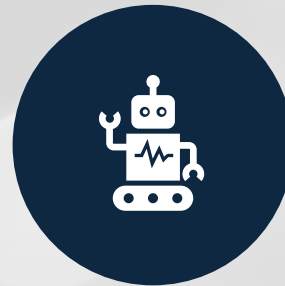
GitLab: DevOps
platform datastore



NASA: telemetry &
research data



Meta/Instagram: high-
scale workloads



Startups: AI, IoT,
analytics

Key Takeaways

PostgreSQL = Secure + Scalable + Extensible

Supports modern workloads (AI, geospatial, hybrid data)

Community-driven innovation

Q&A

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Redgate

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<https://linkedin.com/in/kellyngorman>

